

PACEMAKER & ICD WORKSHOP

SCENE 7 — DEVICES

NBG CODE CHART

DDD for COMPLETE AV BLOCK

Atrial

Ventricular AF

VVI for CHRONIC AF

Atrial

Ventricular AF

PACEMAKER SYNDROME

VVI retrograde conduction

↓

CANNON A WAVES

↓

SOLUTION: UPGRADE TO DDD

Venous pulsations

ICD PRIMARY PREVENTION

- EF ≤ 35%
- NYHA II–III

AOD

WAIT 40D POST-MI / 90D POST-REVASC

ICD SECONDARY PREVENTION

- VT/VF / ABORTED SCD

CRT
(CARDIAC RESYNCHRONIZATION THERAPY)

- EF ≤ 35%
- LBBB
- QRS > 150 ms

→

Pre-resynchronization Post-resynchronization

1. DDD for complete AV block

WHAT YOU SEE

NBG chart: DDD for complete AV block.

WHAT IT ENCODES

A dual-chamber (DDD) pacemaker paces and senses both atrium and ventricle, preserving AV synchrony — preferred for AV block when the atrium is in sinus rhythm. The NBG code = chamber paced / sensed / response.

BOARD TRAP

WRONG: 'a single-chamber ventricular pacemaker is ideal for AV block in sinus rhythm.' DDD preserves AV synchrony and avoids pacemaker syndrome.

2. VVI for chronic AF

WHAT YOU SEE

NBG chart: VVI for chronic AF.

WHAT IT ENCODES

In chronic (permanent) AF there is no organized atrial activity to track, so a single-chamber ventricular (VVI) pacemaker is appropriate — an atrial lead would be useless.

BOARD TRAP

WRONG: 'use DDD in permanent AF.' With no atrial rhythm to sense, dual-chamber pacing adds no benefit — VVI is used.

3. Pacemaker syndrome → upgrade to DDD

WHAT YOU SEE

Pacemaker syndrome: VVI retrograde conduction → upgrade DDD.

WHAT IT ENCODES

Pacemaker syndrome occurs with VVI pacing when retrograde VA conduction causes AV dyssynchrony — fatigue, dyspnea, hypotension, and neck pulsations. Treatment is upgrading to dual-chamber (DDD) pacing.

BOARD TRAP

WRONG: 'pacemaker syndrome means the device has failed and needs replacement.' It's an AV-synchrony problem — upgrade VVI to DDD.

4. Cannon A waves

WHAT YOU SEE

Neck with cannon A waves.

WHAT IT ENCODES

Cannon A waves = the right atrium contracts against a closed tricuspid valve (AV dissociation) — seen in complete heart block, VT, and pacemaker syndrome.

BOARD TRAP

WRONG: 'cannon A waves indicate tricuspid regurgitation.' They reflect atrial contraction against a closed valve from AV dissociation.

5. ICD primary prevention: EF ≤35% + NYHA II–III

WHAT YOU SEE

Bench: ICD primary prevention, EF≤35%, NYHA II-III.

WHAT IT ENCODES

Primary-prevention ICD is indicated for EF ≤35% with NYHA class II–III on optimal medical therapy (ischemic or non-ischemic cardiomyopathy) to prevent sudden cardiac death.

BOARD TRAP

WRONG: 'any low EF qualifies immediately for an ICD.' You must be on optimal medical therapy and past the required waiting period first.

6. Wait 40 d post-MI / 90 d post-revasc

WHAT YOU SEE

Sign: WAIT 40d post-MI / 90d post-revasc.

WHAT IT ENCODES

Re-assess EF and implant a primary-prevention ICD only after ≥40 days post-MI (DINAMIT/IRIS showed no benefit earlier) and ≥90 days post-revascularization — EF often recovers in that window.

BOARD TRAP

WRONG: 'implant the ICD during the index MI admission if EF is low.' Early post-MI ICD shows no mortality benefit — wait and reassess EF.

7. ICD secondary prevention: VT/VF/aborted SCD

WHAT YOU SEE

Bench: secondary prevention, VT/VF/aborted SCD.

WHAT IT ENCODES

Secondary-prevention ICD is for survivors of cardiac arrest (VF/VT) or sustained VT with structural heart disease not from a reversible cause — regardless of EF.

BOARD TRAP

WRONG: 'a normal EF after an aborted cardiac arrest means no ICD.' Survived VF/sustained VT (no reversible cause) warrants an ICD irrespective of EF.

8. CRT: EF ≤35% + LBBB + QRS >150

WHAT YOU SEE

CRT device: EF≤35%, LBBB, QRS>150 ms.

WHAT IT ENCODES

Cardiac resynchronization (biventricular pacing) helps HFrEF with EF ≤35%, LBBB, and wide QRS (greatest benefit ≥150 ms) on optimal therapy — improving symptoms and survival.

BOARD TRAP

WRONG: 'CRT helps any wide QRS.' Benefit is greatest with LBBB and QRS ≥150 ms; non-LBBB/narrower QRS benefit little.